

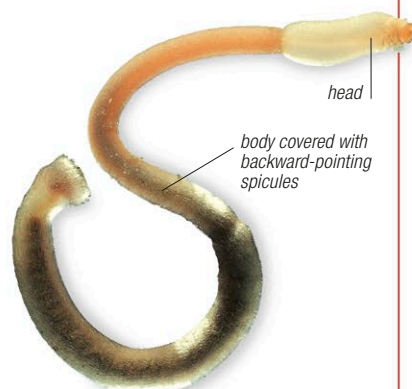
Class Aplacophora

Aplacophorans

Occurrence 395 spp. worldwide; on the seabed or in association with other animals (especially cnidarians)



With their thin, wormlike bodies, aplacophorans bear very little resemblance to any other mollusks. Typically less than $\frac{3}{16}$ in (5 mm) long, they have no trace of a shell, but are covered by a tough cuticle that contains a large number of small, mineral spicules. They have poorly developed heads and no eyes, and move by creeping or by burrowing through sediment. The creeping species feed on cnidarians, while the burrowers feed on tiny animals, or on particles of organic matter. Some species have separate sexes, but most are hermaphrodites. Some keep their eggs until they hatch, but little else is known about their breeding habits.

**CHAETODERMA SPP.**

Members of the genus *Chaetoderma*, found in the Northern Hemisphere, are relative giants among aplacophorans, growing up to $\frac{3}{4}$ in (8 cm) long. Like other aplacophorans, they lack a foot. They breathe with gills, located in a cavity at the back of the body.

Class Polyplacophora

Chitons

Occurrence 950 spp. worldwide; on the seabed



Chitons resemble gastropods in having a suckerlike foot, but their shells are quite distinctive, being made of 8 plates. These are arranged in a line, and they arch over the animal's back. These fat, oval to elongated mollusks are covered by a thick "girdle," which often flanks the plates like a lip. It has a rubbery texture, and in many species is covered with bristles or calcareous spicules. Chitons can be up to 16 in (40 cm) long. They have a large, muscular foot, a small head with a well-developed rasping mouthpart (radula), but no eyes or tentacles, and a U-shaped mantle cavity around the foot with 6–88 pairs of gills. In both sexes, there is one pair of gonads (reproductive organs) fused into

a single structure. Chitons graze algae from rocks on the seabed. They creep slowly over the surface—often after dark or at high tide—and return to a

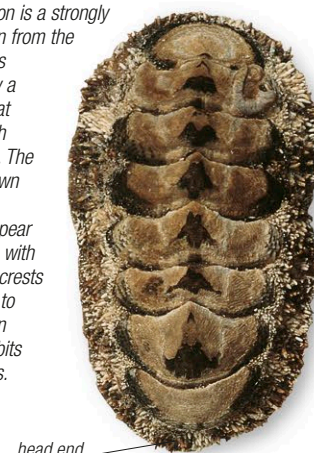
particular site where they fit snugly against the rock, helping to protect themselves from attack, and from desiccation if exposed to the air.

**CHITON STRIATUS**

Found on the west coast of South America, the magnificent chiton is large and well arched and surrounded by a broad girdle. It measures $\frac{3}{2}$ in (9 cm) in length and inhabits intertidal rocks. The central portion of each plate has crowded, straight riblets that are usually worn smooth at the highest point. Head and tail plates have coarse, radiating ribs, and the girdle is covered with polished granules.

ACANTHOPLEURA GRANULATA

The fuzzy chiton is a strongly humped chiton from the Caribbean. It is surrounded by a thick girdle that is covered with coarse spines. The plates are brown when unworn, but usually appear grayish brown with darker brown crests and sides. Up to $2\frac{1}{4}$ in (6 cm) in length, it inhabits intertidal rocks.

**MARbled CHITON**

Tonicella marmorea, a North Atlantic chiton up to $1\frac{1}{2}$ in (4 cm) long, has a broad, leathery girdle surrounding the shell plates. It can be found from the intertidal zone to depths of 650 ft (200 m). It occurs on rocks, stones, and pebbles, and in some Scottish sea lochs can attain densities of 50 per square yard.



Class Monoplacophora

Monoplacophorans

Occurrence 30 spp. in Atlantic, Pacific, and Indian oceans; at depths of 650–23,100 ft+ (200–7,000 m)



Monoplacophoran mollusks were thought to have become extinct more than 300 million years ago, until a handful of specimens were dredged up in the eastern Pacific Ocean in 1952. Since then, they have been found on the deep seabed in widely scattered parts of the world. Monoplacophorans look like limpets (see p.596), but they each have up to 8 pairs of foot retractor muscles, whereas limpets and other gastropods have just one pair. A horseshoe-shaped mantle cavity containing 3–6 pairs of gills surrounds the foot. The shell is conical or cap-shaped, and the head is small, with a large mouthpart (radula). There are no eyes or tentacles. Sexes are separate. These mollusks grow up to $1\frac{1}{4}$ in (3 cm) long.

Class Scaphopoda

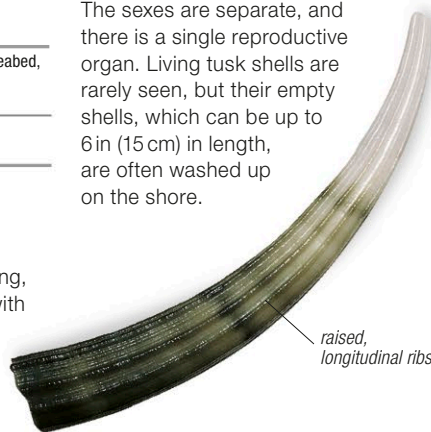
Tusk shells

Occurrence About 580 spp. worldwide; on the seabed, in soft sediments



Widely distributed in all the world's oceans, these burrowing mollusks are easily recognized by the shape of their shells, which are like tapering, hollow tusks, open at both ends, with one opening larger than the other. Tusk shells live head downward in sediment, the head and the foot protruding out of the larger of the shell openings. The small head has numerous fine, ciliated feeding filaments, which contract to transport food particles from the sediment into the mouth. The smaller aperture projects above the sediment surface; respiratory water current is drawn in and let out through this opening. There are no gills; instead, gases are exchanged through the mantle surface.

The sexes are separate, and there is a single reproductive organ. Living tusk shells are rarely seen, but their empty shells, which can be up to 6 in (15 cm) in length, are often washed up on the shore.

**ELEPHANT TUSK**

The solid and slightly glossy shell of the species *Dentalium elephantinum* has a characteristic notch at its narrower end. This mollusk inhabits offshore sands in the Indo-Pacific, and is 3 in (7.5 cm) long.

ELONGATED TUSK

Antalis species are a small group of marine mollusks from the North Atlantic that live in shells $2\frac{1}{4}$ in (6 cm) long. They feed on microorganisms, using numerous club-shaped tentacles.

**BEAUTIFUL TUSK**

This species, *Pictodentalium formosum*, found on Indo-Pacific coasts, grows to 3 in (7.5 cm) in length. It differs from other large tusk shells in its head end being only slightly broader than its rear end; it is also straighter than most. Its longitudinal ribs and riblets are crossed by concentric growth lines. A small, notched pipe protrudes from the rear end.